

Baseline

Dataset001_BrainMets

Task: Multi-class tumor segmentation (necrotic core, enhancing tumor, edema) on brain MRI (T1, T1c, T2, FLAIR) + primary tumor origin classification (binary: NSCLC vs Breast carcinoma)

Note: The classification head outputs 3 classes (0: NSCLC, 1: Breast carcinoma, 2: Other origin – present in the full training cohort but excluded from this public dataset). Since no label-2 cases appear in the public split, AUROC/AUPRC are computed as binary using the class-1 (Breast carcinoma) probability against class-0 (NSCLC).

Per-Fold Results – Segmentation

Fold	DSC (necrotic)	DSC (enhancing)	DSC (edema)	DSC mean	NSD (necrotic)	NSD (enhancing)	NSD (edema)	NSD mean
0	0.693	0.541	0.633	0.622	0.761	0.659	0.710	0.710
1	0.634	0.605	0.485	0.575	0.719	0.732	0.602	0.684
2	0.591	0.664	0.544	0.599	0.679	0.828	0.733	0.746
3	0.589	0.553	0.399	0.514	0.629	0.669	0.527	0.608
4	0.621	0.601	0.633	0.619	0.713	0.699	0.697	0.703
Mean	0.626	0.593	0.539	0.586	0.700	0.717	0.654	0.690
Std	0.042	0.049	0.100	0.045	0.050	0.068	0.087	0.051

Per-Fold Results – Classification

Fold	AUROC	AUPRC	Bal Acc	F1
0	0.849	0.824	0.764	0.776
1	0.848	0.840	0.772	0.766
2	0.938	0.867	0.840	0.771
3	1.000	1.000	0.955	0.921
4	0.993	0.994	0.611	0.530
Mean	0.926	0.905	0.788	0.753
Std	0.074	0.085	0.125	0.141

Figures

![Box plots](figures/01_boxplot_dsc_nsd.png)

! [Case sampling](figures/02_case_sampling.png)
! [ROC/PRC](figures/03_roc_prc.png)
! [Confusion matrix](figures/04_confusion_matrix.png)
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Dataset002_MU_Glioma

Task: Multi-class glioma segmentation (NCR, ED, ET, NET_RC) on brain MRI (T1c, T1, FLAIR, T2) + primary diagnosis classification (3-class: GBM, Astrocytoma, Others)

Per-Fold Results – Segmentation

Fold	DSC (NCR)	DSC (ED)	DSC (ET)	DSC (NET_RC)	DSC mean	NSD (NCR)	NSD (ED)	NSD (ET)	NSD (NET_RC)	NSD mean
0	0.650	0.850	0.764	0.547	0.703	0.697	0.896	0.834	0.621	0.762
1	0.623	0.888	0.794	0.648	0.738	0.678	0.931	0.865	0.711	0.796
2	0.621	0.906	0.734	0.622	0.721	0.659	0.944	0.822	0.677	0.775
3	0.605	0.870	0.822	0.583	0.720	0.661	0.908	0.894	0.635	0.775
4	0.623	0.860	0.755	0.551	0.697	0.706	0.918	0.834	0.603	0.765
Mean	0.625	0.875	0.774	0.590	0.716	0.680	0.919	0.850	0.649	0.775
Std	0.016	0.023	0.034	0.044	0.016	0.021	0.019	0.029	0.044	0.013

Per-Fold Results – Classification

Fold	AUROC	AUPRC	Bal Acc	F1
0	0.882	0.610	0.599	0.562
1	0.879	0.651	0.660	0.631
2	0.910	0.650	0.540	0.514
3	0.894	0.663	0.640	0.615
4	0.920	0.591	0.505	0.513
Mean	0.897	0.633	0.589	0.567
Std	0.018	0.031	0.065	0.055

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Dataset003_UCSD_PTGB

Task: Tumor segmentation on brain MRI (T1c, FLAIR, ADC) + IDH mutation status classification (binary: Wild-Type vs Mutant)

Per-Fold Results

Fold	DSC (tumor)	NSD (tumor)	AUROC	AUPRC	Bal Acc	F1
0	0.535	0.592	0.768	0.205	0.500	0.000
1	0.399	0.500	0.615	0.141	0.661	0.250
2	0.406	0.481	0.721	0.235	0.457	0.000
3	0.558	0.625	0.667	0.140	0.800	0.300
4	0.463	0.524	0.618	0.102	0.794	0.222
Mean	0.472	0.544	0.678	0.165	0.642	0.154
Std	0.072	0.062	0.067	0.054	0.160	0.144

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Dataset004_PICAI

Task: Multi-class prostate segmentation (prostate gland + tumor) on prostate MRI (T2W, ADC, HBV) + ISUP grade classification (6-class: 0=Benign/Indolent, 1-5=ISUP grades)

Per-Fold Results – Segmentation

Fold	DSC (prostate)	NSD (prostate)	DSC (tumor)	NSD (tumor)	DSC mean	NSD mean
0	0.960	0.959	0.473	0.490	0.717	0.724
1	0.960	0.960	0.536	0.552	0.748	0.756
2	0.959	0.959	0.634	0.649	0.797	0.804
3	0.947	0.939	0.168	0.188	0.558	0.564
4	0.960	0.960	0.557	0.575	0.759	0.767
Mean	**0.957**	**0.955**	**0.474**	**0.491**	**0.716**	**0.723**
Std	**0.006**	**0.009**	**0.180**	**0.179**	**0.093**	**0.094**

Per-Fold Results – Classification

Fold	AUROC	AUPRC	Bal Acc	F1
0	0.731	0.290	0.240	0.236
1	0.720	0.281	0.212	0.183
2	0.746	0.300	0.240	0.229
3	0.702	0.310	0.176	0.063
4	0.736	0.350	0.263	0.224
Mean	**0.727**	**0.306**	**0.226**	**0.187**
Std	**0.017**	**0.027**	**0.033**	**0.072**

Notes:

- Prostate gland segmentation is strong and consistent (DSC 0.957 ± 0.006); the gland is a well-defined, large structure
- Tumor segmentation is challenging (DSC 0.474 ± 0.180): 71% of cases (1075/1500) have no tumor, so many folds see near-zero tumor Dice dominated by true-negatives
- Fold 3 has noticeably lower tumor DSC (0.168) and NSD (0.188), likely due to an unfavorable fold split or harder cases
- Classification: AUROC 0.727 is reasonable for 6-class ISUP grading; Bal Acc (0.226) and F1 (0.187) are near the random baseline ($1/6 \approx 0.167$), driven by severe class imbalance (847/1500 cases are grade 0)

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Dataset005_LUNA25

Task: Lung nodule segmentation on chest CT + malignancy classification (binary)

Per-Fold Results

Fold	DSC (nodule)	NSD (nodule)	AUROC	AUPRC	Bal Acc	F1
0	0.751	0.902	0.819	0.397	0.730	0.295
1	0.724	0.886	0.900	0.567	0.719	0.545
2	0.749	0.911	0.880	0.509	0.802	0.513
3	0.732	0.892	0.884	0.531	0.807	0.444
4	0.735	0.895	0.867	0.440	0.804	0.470
Mean	**0.738**	**0.897**	**0.870**	**0.489**	**0.772**	**0.453**
Std	**0.012**	**0.010**	**0.031**	**0.069**	**0.044**	**0.096**

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Dataset061_PETWB_Lung

Task: Lung segmentation on whole-body PET/CT + lung cancer classification (binary)

Per-Fold Results

Fold	DSC (lung)	NSD (lung)	AUROC	AUPRC	Bal Acc	F1
0	0.981	0.915	0.638	0.954	0.494	0.952
1	0.984	0.929	0.672	0.961	0.660	0.611
2	0.972	0.882	0.758	0.975	0.500	0.000

	3		0.980		0.918		0.586		0.936		0.500		0.952	
	4		0.978		0.920		0.748		0.961		0.500		0.952	
	Mean		**0.979**		**0.913**		**0.680**		**0.957**		**0.531**			
	0.693													
	Std		**0.004**		**0.018**		**0.073**		**0.014**		**0.072**			
	0.415													

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Dataset062_PETWB_Liver

Task: Liver segmentation on whole-body PET/CT + liver cancer classification (binary)

Per-Fold Results

Fold	DSC (liver)	NSD (liver)	AUROC	AUPRC	Bal Acc	F1
0	0.965	0.849	0.648	0.609	0.500	0.667
1	0.965	0.861	0.666	0.741	0.653	0.541
2	0.956	0.833	0.680	0.724	0.640	0.696
3	0.962	0.846	0.648	0.735	0.648	0.533
4	0.960	0.836	0.663	0.693	0.599	0.661
Mean	**0.962**	**0.845**	**0.661**	**0.700**	**0.608**	
0.619						
Std	**0.004**	**0.011**	**0.014**	**0.055**	**0.064**	
0.077						

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Dataset001 BrainMets

Rank	Model	DSC	NSD	Bal. Acc.	AUROC	Mean Rank
1	A3	0.8156	0.9090	0.6321	0.5184	2.25
2	A1	0.8156	0.9090	0.6321	0.4649	2.50
2	A2	0.8156	0.9090	0.6104	0.5920	2.50
4	A0	0.8156	0.9090	0.5000	0.1438	3.50
4	A4	0.8110	0.9027	0.6538	0.6288	3.50
4	A5	0.8147	0.9078	0.6488	0.6120	3.50

Dataset003 UCSD_PTGB

Rank	Model	DSC	NSD	Bal. Acc.	AUROC	Mean Rank
1	A0	0.9222	0.9645	0.6364	0.6061	1.50
2	A3	0.9222	0.9645	0.6364	0.3434	1.75
3	A2	0.9222	0.9645	0.5000	0.3434	2.25
4	A1	0.9222	0.9645	0.4848	0.3131	3.50
4	A4	0.8325	0.8675	0.5000	0.8788	3.50
4	A5	0.8325	0.8675	0.5000	0.8788	3.50

Dataset004 PICAI

Rank	Model	DSC	NSD	Bal. Acc.	AUROC	Mean Rank
1	A5	0.6295	0.6131	0.3025	0.8067	2.00
2	A1	0.6253	0.6080	0.3064	0.7383	2.50
3	A3	0.6253	0.6080	0.3149	0.7248	2.75
4	A2	0.6253	0.6080	0.2907	0.7370	3.50
4	A4	0.5900	0.5690	0.3223	0.8122	3.50
6	A0	0.6253	0.6080	0.2985	0.6738	3.75

Dataset005 LUNA25

Rank	Model	DSC	NSD	Bal. Acc.	AUROC	Mean Rank
1	A3	0.7722	0.8621	0.7923	0.8422	1.50
2	A1	0.7722	0.8621	0.7812	0.8312	2.25
2	A5	0.7837	0.8685	0.7177	0.8421	2.25
4	A2	0.7722	0.8621	0.7750	0.8259	3.00
5	A0	0.7722	0.8621	0.7580	0.8099	3.50
6	A4	0.7422	0.8274	0.7021	0.8282	5.50

Dataset061 PETWB_Lung

Rank	Model	DSC	NSD	Bal. Acc.	AUROC	Mean Rank
1	A2	0.9879	0.9413	0.7472	0.7417	1.00
2	A3	0.9879	0.9413	0.6472	0.6569	2.00
3	A5	0.9879	0.9413	0.5000	0.6597	2.25
4	A0	0.9879	0.9413	0.6653	0.6542	2.50
4	A1	0.9879	0.9413	0.6403	0.6556	2.50
6	A4	0.9879	0.9413	0.5000	0.6556	2.75

Dataset062 PETWB_Liver

Rank	Model	DSC	NSD	Bal. Acc.	AUROC	Mean Rank
1	A1	0.9786	0.9006	0.5918	0.6031	1.25
1	A3	0.9786	0.9006	0.6020	0.5706	1.25
3	A5	0.9786	0.9006	0.5816	0.5494	2.25
4	A2	0.9786	0.9006	0.5714	0.5589	2.50
4	A4	0.9786	0.9006	0.5816	0.5410	2.50
6	A0	0.9786	0.9006	0.5612	0.5352	3.50

Dataset001_BrainMets

A0

AutoMSC Evaluation

--- Segmentation Metrics ---

Label 1: DSC = 0.7690 +/- 0.1659 | NSD = 0.8675 +/- 0.1206

Label 2: DSC = 0.8206 +/- 0.0928 | NSD = 0.9224 +/- 0.0801

Label 3: DSC = 0.8368 +/- 0.1386 | NSD = 0.9171 +/- 0.0734

Overall: DSC = 0.8156 +/- 0.1285 | NSD = 0.9090 +/- 0.0894

--- Classification Metrics ---

Accuracy: 0.3611

Balanced Accuracy: 0.5000

AUC: 0.1438

AUPRC: 0.2411

Sensitivity: 1.0000

Specificity: 0.0000

Precision: 0.3611

Recall: 1.0000

F1: 0.5306

Confusion Matrix:

[[0 23]

[0 13]]

--- Balanced Accuracy Verification ---

Definition: unweighted mean of per-class recall

class 0: recall=0.0000, n=23 | class 1: recall=1.0000, n=13

Verified Balanced Accuracy: 0.5000

--- Efficiency Metrics ---

Mean inference time: 7.1141 s

Maximum inference time: 35.4815 s

Mean peak GPU memory: 2216.77 MB

Maximum peak GPU memory: 3310.81 MB

A1

AutoMSC Evaluation

--- Segmentation Metrics ---

Label 1: DSC = 0.7690 +/- 0.1659 | NSD = 0.8675 +/- 0.1206

Label 2: DSC = 0.8206 +/- 0.0928 | NSD = 0.9224 +/- 0.0801

Label 3: DSC = 0.8368 +/- 0.1386 | NSD = 0.9171 +/- 0.0734

Overall: DSC = 0.8156 +/- 0.1285 | NSD = 0.9090 +/- 0.0894

--- Classification Metrics ---

Accuracy: 0.7222

Balanced Accuracy: 0.6321

AUC: 0.4649

AUPRC: 0.5539

Sensitivity: 0.3077

Specificity: 0.9565

Precision: 0.8000

Recall: 0.3077

F1: 0.4444

Confusion Matrix:

[[22 1]

[9 4]]

--- Balanced Accuracy Verification ---

Definition: unweighted mean of per-class recall

class 0: recall=0.9565, n=23 | class 1: recall=0.3077, n=13

Verified Balanced Accuracy: 0.6321

--- Efficiency Metrics ---

Mean inference time: 14.1462 s

Maximum inference time: 44.1369 s

Mean peak GPU memory: 2218.33 MB

Maximum peak GPU memory: 3312.37 MB

A2

AutoMSC Evaluation

--- Segmentation Metrics ---

Label 1: DSC = 0.7690 +/- 0.1659 | NSD = 0.8675 +/- 0.1206

Label 2: DSC = 0.8206 +/- 0.0928 | NSD = 0.9224 +/- 0.0801

Label 3: DSC = 0.8368 +/- 0.1386 | NSD = 0.9171 +/- 0.0734

Overall: DSC = 0.8156 +/- 0.1285 | NSD = 0.9090 +/- 0.0894

--- Classification Metrics ---

Accuracy: 0.6944

Balanced Accuracy: 0.6104

AUC: 0.5920

AUPRC: 0.6002

Sensitivity: 0.3077

Specificity: 0.9130

Precision: 0.6667

Recall: 0.3077

F1: 0.4211

Confusion Matrix:

[[21 2]

[9 4]]

--- Balanced Accuracy Verification ---

Definition: unweighted mean of per-class recall

class 0: recall=0.9130, n=23 | class 1: recall=0.3077, n=13

Verified Balanced Accuracy: 0.6104

--- Efficiency Metrics ---

Mean inference time: 4.1800 s

Maximum inference time: 5.7195 s

Mean peak GPU memory: 2576.76 MB

Maximum peak GPU memory: 2578.32 MB

A3

AutoMSC Evaluation

--- Segmentation Metrics ---

Label 1: DSC = 0.7690 +/- 0.1659 | NSD = 0.8675 +/- 0.1206

Label 2: DSC = 0.8206 +/- 0.0928 | NSD = 0.9224 +/- 0.0801

Label 3: DSC = 0.8368 +/- 0.1386 | NSD = 0.9171 +/- 0.0734

Overall: DSC = 0.8156 +/- 0.1285 | NSD = 0.9090 +/- 0.0894

--- Classification Metrics ---

Accuracy: 0.7222

Balanced Accuracy: 0.6321

AUC: 0.5184

AUPRC: 0.5613

Sensitivity: 0.3077

Specificity: 0.9565

Precision: 0.8000

Recall: 0.3077

F1: 0.4444

Confusion Matrix:

[[22 1]

[9 4]]

--- Balanced Accuracy Verification ---

Definition: unweighted mean of per-class recall

class 0: recall=0.9565, n=23 | class 1: recall=0.3077, n=13

Verified Balanced Accuracy: 0.6321

--- Efficiency Metrics ---

Mean inference time: 4.1696 s

Maximum inference time: 5.5439 s

Mean peak GPU memory: 2968.95 MB

Maximum peak GPU memory: 2970.52 MB

A4

AutoMSC Evaluation

--- Segmentation Metrics ---

Label 1: DSC = 0.7701 +/- 0.1651 | NSD = 0.8677 +/- 0.1212

Label 2: DSC = 0.8182 +/- 0.0934 | NSD = 0.9176 +/- 0.0810

Label 3: DSC = 0.8261 +/- 0.1380 | NSD = 0.9052 +/- 0.0806

Overall: DSC = 0.8110 +/- 0.1276 | NSD = 0.9027 +/- 0.0914

--- Classification Metrics ---

Accuracy: 0.7500

Balanced Accuracy: 0.6538

AUC: 0.6288

AUPRC: 0.6308

Sensitivity: 0.3077

Specificity: 1.0000

Precision: 1.0000

Recall: 0.3077

F1: 0.4706

Confusion Matrix:

[[23 0]

[9 4]]

--- Balanced Accuracy Verification ---

Definition: unweighted mean of per-class recall

class 0: recall=1.0000, n=23 | class 1: recall=0.3077, n=13

Verified Balanced Accuracy: 0.6538

--- Efficiency Metrics ---

Mean inference time: 5.3779 s

Maximum inference time: 26.0985 s

Mean peak GPU memory: 6335.10 MB

Maximum peak GPU memory: 6336.68 MB

A5

AutoMSC Evaluation

--- Segmentation Metrics ---

Label 1: DSC = 0.7693 +/- 0.1652 | NSD = 0.8668 +/- 0.1213

Label 2: DSC = 0.8196 +/- 0.0919 | NSD = 0.9205 +/- 0.0807

Label 3: DSC = 0.8354 +/- 0.1407 | NSD = 0.9165 +/- 0.0740

Overall: DSC = 0.8147 +/- 0.1288 | NSD = 0.9078 +/- 0.0899

--- Classification Metrics ---

Accuracy: 0.7222

Balanced Accuracy: 0.6488

AUC: 0.6120

AUPRC: 0.5309

Sensitivity: 0.3846

Specificity: 0.9130

Precision: 0.7143

Recall: 0.3846

F1: 0.5000

Confusion Matrix:

[[21 2]

[8 5]]

--- Balanced Accuracy Verification ---

Definition: unweighted mean of per-class recall

class 0: recall=0.9130, n=23 | class 1: recall=0.3846, n=13

Verified Balanced Accuracy: 0.6488

--- Efficiency Metrics ---

Mean inference time: 5.3400 s

Maximum inference time: 25.8525 s

Mean peak GPU memory: 4404.47 MB

Maximum peak GPU memory: 4406.03 MB

Dataset002_MU-Glioma_Post

A0

AutoMSC Evaluation

--- Segmentation Metrics ---

Label 1: DSC = 0.6584 +/- 0.3791 | NSD = 0.7084 +/- 0.3791

Label 2: DSC = 0.9225 +/- 0.0955 | NSD = 0.9500 +/- 0.0881

Label 3: DSC = 0.7848 +/- 0.3118 | NSD = 0.8431 +/- 0.3195

Label 4: DSC = 0.7305 +/- 0.3108 | NSD = 0.7843 +/- 0.2977

Overall: DSC = 0.7900 +/- 0.2944 | NSD = 0.8368 +/- 0.2893

--- Classification Metrics ---

Accuracy: 0.7379

Balanced Accuracy: 0.5294

Balanced Accuracy: 0.5294

Weighted AUC: 0.7497

Weighted AUPRC: 0.7411

Weighted F1: 0.7407

Weighted Precision: 0.7755

Weighted Recall: 0.7379

Per-class metrics:

Class 0: Precision=0.9286, Recall=0.8025, F1=0.8609, AUC=0.7744, AUPRC=0.8593

Class 1: Precision=0.3333, Recall=0.7857, F1=0.4681, AUC=0.7055, AUPRC=0.4275

Class 2: Precision=0.0000, Recall=0.0000, F1=0.0000, AUC=0.5763, AUPRC=0.0936

Confusion Matrix:

[[65 16 0]

[3 11 0]

[2 6 0]]

--- Balanced Accuracy Verification ---

Definition: unweighted mean of per-class recall

class 0: recall=0.8025, n=81 | class 1: recall=0.7857, n=14 | class 2: recall=0.0000, n=8

Verified Balanced Accuracy: 0.5294

--- Efficiency Metrics ---

Mean inference time: 8.7372 s

Maximum inference time: 385.8481 s

Mean peak GPU memory: 1700.45 MB

Maximum peak GPU memory: 2670.74 MB

A1

AutoMSC Evaluation

--- Segmentation Metrics ---

Label 1: DSC = 0.6584 +/- 0.3791 | NSD = 0.7084 +/- 0.3791

Label 2: DSC = 0.9225 +/- 0.0955 | NSD = 0.9500 +/- 0.0881

Label 3: DSC = 0.7848 +/- 0.3118 | NSD = 0.8431 +/- 0.3195

Label 4: DSC = 0.7305 +/- 0.3108 | NSD = 0.7843 +/- 0.2977

Overall: DSC = 0.7900 +/- 0.2944 | NSD = 0.8368 +/- 0.2893

--- Classification Metrics ---

Accuracy: 0.8155

Balanced Accuracy: 0.6553

Balanced Accuracy: 0.6553

Weighted AUC: 0.7818

Weighted AUPRC: 0.7604

Weighted F1: 0.8202

Weighted Precision: 0.8300

Weighted Recall: 0.8155

Per-class metrics:

Class 0: Precision=0.9221, Recall=0.8765, F1=0.8987, AUC=0.7952, AUPRC=0.8656

Class 1: Precision=0.5263, Recall=0.7143, F1=0.6061, AUC=0.7271, AUPRC=0.4170

Class 2: Precision=0.4286, Recall=0.3750, F1=0.4000, AUC=0.7421, AUPRC=0.2959

Confusion Matrix:

[[71 6 4]

[4 10 0]

[2 3 3]]

--- Balanced Accuracy Verification ---

Definition: unweighted mean of per-class recall

class 0: recall=0.8765, n=81 | class 1: recall=0.7143, n=14 | class 2: recall=0.3750, n=8

Verified Balanced Accuracy: 0.6553

--- Efficiency Metrics ---

Mean inference time: 3.6164 s

Maximum inference time: 3.9647 s

Mean peak GPU memory: 2084.26 MB

Maximum peak GPU memory: 2102.84 MB

A2

AutoMSC Evaluation

--- Segmentation Metrics ---

Label 1: DSC = 0.6584 +/- 0.3791 | NSD = 0.7084 +/- 0.3791
Label 2: DSC = 0.9225 +/- 0.0955 | NSD = 0.9500 +/- 0.0881
Label 3: DSC = 0.7848 +/- 0.3118 | NSD = 0.8431 +/- 0.3195
Label 4: DSC = 0.7305 +/- 0.3108 | NSD = 0.7843 +/- 0.2977
Overall: DSC = 0.7900 +/- 0.2944 | NSD = 0.8368 +/- 0.2893

--- Classification Metrics ---

Accuracy: 0.6796
Balanced Accuracy: 0.6352
Balanced Accuracy: 0.6352
Weighted AUC: 0.7044
Weighted AUPRC: 0.7519
Weighted F1: 0.7052
Weighted Precision: 0.7660
Weighted Recall: 0.6796

Per-class metrics:

Class 0: Precision=0.8750, Recall=0.6914, F1=0.7724, AUC=0.7059, AUPRC=0.8570
Class 1: Precision=0.3448, Recall=0.7143, F1=0.4651, AUC=0.7151, AUPRC=0.3685
Class 2: Precision=0.4000, Recall=0.5000, F1=0.4444, AUC=0.6697, AUPRC=0.3582

Confusion Matrix:

```
[[56 19 6]
 [ 4 10 0]
 [ 4 0 4]]
```

--- Balanced Accuracy Verification ---

Definition: unweighted mean of per-class recall
class 0: recall=0.6914, n=81 | class 1: recall=0.7143, n=14 | class 2: recall=0.5000, n=8
Verified Balanced Accuracy: 0.6352

--- Efficiency Metrics ---

Mean inference time: 3.6146 s
Maximum inference time: 3.9815 s
Mean peak GPU memory: 2475.12 MB
Maximum peak GPU memory: 2494.62 MB

A3

AutoMSC Evaluation

--- Segmentation Metrics ---

Label 1: DSC = 0.6584 +/- 0.3791 | NSD = 0.7084 +/- 0.3791
Label 2: DSC = 0.9225 +/- 0.0955 | NSD = 0.9500 +/- 0.0881
Label 3: DSC = 0.7848 +/- 0.3118 | NSD = 0.8431 +/- 0.3195
Label 4: DSC = 0.7305 +/- 0.3108 | NSD = 0.7843 +/- 0.2977
Overall: DSC = 0.7900 +/- 0.2944 | NSD = 0.8368 +/- 0.2893

--- Classification Metrics ---

Accuracy: 0.7961
Balanced Accuracy: 0.6095
Balanced Accuracy: 0.6095
Weighted AUC: 0.7408
Weighted AUPRC: 0.7751
Weighted F1: 0.7940
Weighted Precision: 0.8036
Weighted Recall: 0.7961

Per-class metrics:

Class 0: Precision=0.8861, Recall=0.8642, F1=0.8750, AUC=0.7531, AUPRC=0.8495
Class 1: Precision=0.5000, Recall=0.7143, F1=0.5882, AUC=0.7536, AUPRC=0.6034
Class 2: Precision=0.5000, Recall=0.2500, F1=0.3333, AUC=0.5934, AUPRC=0.3218

Confusion Matrix:

```
[[70 9 2]
 [ 4 10 0]
 [ 5 1 2]]
```

--- Balanced Accuracy Verification ---

Definition: unweighted mean of per-class recall
class 0: recall=0.8642, n=81 | class 1: recall=0.7143, n=14 | class 2: recall=0.2500, n=8
Verified Balanced Accuracy: 0.6095

--- Efficiency Metrics ---

Mean inference time: 3.6244 s
Maximum inference time: 3.9971 s
Mean peak GPU memory: 2868.74 MB
Maximum peak GPU memory: 2888.58 MB

A4

AutoMSC Evaluation

--- Segmentation Metrics ---

Label 1: DSC = 0.3762 +/- 0.4000 | NSD = 0.4086 +/- 0.4049

Label 2: DSC = 0.7946 +/- 0.1958 | NSD = 0.8007 +/- 0.1881

Label 3: DSC = 0.6721 +/- 0.3389 | NSD = 0.7328 +/- 0.3470

Label 4: DSC = 0.4731 +/- 0.3454 | NSD = 0.5108 +/- 0.3569

Overall: DSC = 0.5939 +/- 0.3595 | NSD = 0.6280 +/- 0.3621

--- Classification Metrics ---

Accuracy: 0.8641

Balanced Accuracy: 0.6365

Balanced Accuracy: 0.6365

Weighted AUC: 0.7955

Weighted AUPRC: 0.7932

Weighted F1: 0.8571

Weighted Precision: 0.8538

Weighted Recall: 0.8641

Per-class metrics:

Class 0: Precision=0.9176, Recall=0.9630, F1=0.9398, AUC=0.8072, AUPRC=0.8906

Class 1: Precision=0.7273, Recall=0.5714, F1=0.6400, AUC=0.7448, AUPRC=0.5217

Class 2: Precision=0.4286, Recall=0.3750, F1=0.4000, AUC=0.7658, AUPRC=0.2829

Confusion Matrix:

[[78 2 1]

[3 8 3]

[4 1 3]]

--- Balanced Accuracy Verification ---

Definition: unweighted mean of per-class recall

class 0: recall=0.9630, n=81 | class 1: recall=0.5714, n=14 | class 2: recall=0.3750, n=8

Verified Balanced Accuracy: 0.6365

--- Efficiency Metrics ---

Mean inference time: 3.6747 s

Maximum inference time: 4.2393 s

Mean peak GPU memory: 5714.19 MB

Maximum peak GPU memory: 5734.11 MB

A5

AutoMSC Evaluation

--- Segmentation Metrics ---

Label 1: DSC = 0.6624 +/- 0.3767 | NSD = 0.7081 +/- 0.3845

Label 2: DSC = 0.9180 +/- 0.1011 | NSD = 0.9454 +/- 0.0911

Label 3: DSC = 0.7748 +/- 0.3145 | NSD = 0.8353 +/- 0.3213

Label 4: DSC = 0.7449 +/- 0.3065 | NSD = 0.7916 +/- 0.3095

Overall: DSC = 0.7908 +/- 0.2925 | NSD = 0.8355 +/- 0.2937

--- Classification Metrics ---

Accuracy: 0.8641

Balanced Accuracy: 0.5829

Balanced Accuracy: 0.5829

Weighted AUC: 0.7837

Weighted AUPRC: 0.7800

Weighted F1: 0.8341

Weighted Precision: 0.8089

Weighted Recall: 0.8641

Per-class metrics:

Class 0: Precision=0.9286, Recall=0.9630, F1=0.9455, AUC=0.7924, AUPRC=0.8638

Class 1: Precision=0.5789, Recall=0.7857, F1=0.6667, AUC=0.7568, AUPRC=0.5724

Class 2: Precision=0.0000, Recall=0.0000, F1=0.0000, AUC=0.7434, AUPRC=0.2950

Confusion Matrix:

[[78 3 0]

[3 11 0]

[3 5 0]]

--- Balanced Accuracy Verification ---

Definition: unweighted mean of per-class recall

class 0: recall=0.9630, n=81 | class 1: recall=0.7857, n=14 | class 2: recall=0.0000, n=8

Verified Balanced Accuracy: 0.5829

--- Efficiency Metrics ---

Mean inference time: 4.2091 s

Maximum inference time: 30.4645 s

Mean peak GPU memory: 6437.89 MB

Maximum peak GPU memory: 6457.49 MB

Dataset003_UCSD_PTGB

A0

AutoMSC Evaluation

--- Segmentation Metrics ---

Label 1: DSC = 0.9222 +/- 0.0370 | NSD = 0.9645 +/- 0.0249

Overall: DSC = 0.9222 +/- 0.0370 | NSD = 0.9645 +/- 0.0249

--- Classification Metrics ---

Accuracy: 0.3333

Balanced Accuracy: 0.6364

AUC: 0.6061

AUPRC: 0.1292

Sensitivity: 1.0000

Specificity: 0.2727

Precision: 0.1111

Recall: 1.0000

F1: 0.2000

Confusion Matrix:

[[9 24]

[0 3]]

--- Balanced Accuracy Verification ---

Definition: unweighted mean of per-class recall

class 0: recall=0.2727, n=33 | class 1: recall=1.0000, n=3

Verified Balanced Accuracy: 0.6364

--- Efficiency Metrics ---

Mean inference time: 15.5065 s

Maximum inference time: 49.4721 s

Mean peak GPU memory: 1659.76 MB

Maximum peak GPU memory: 3158.06 MB

A1

AutoMSC Evaluation

--- Segmentation Metrics ---

Label 1: DSC = 0.9222 +/- 0.0370 | NSD = 0.9645 +/- 0.0249

Overall: DSC = 0.9222 +/- 0.0370 | NSD = 0.9645 +/- 0.0249

--- Classification Metrics ---

Accuracy: 0.8889

Balanced Accuracy: 0.4848

AUC: 0.3131

AUPRC: 0.1148

Sensitivity: 0.0000

Specificity: 0.9697

Precision: 0.0000

Recall: 0.0000

F1: 0.0000

Confusion Matrix:

[[32 1]

[3 0]]

--- Balanced Accuracy Verification ---

Definition: unweighted mean of per-class recall

class 0: recall=0.9697, n=33 | class 1: recall=0.0000, n=3

Verified Balanced Accuracy: 0.4848

--- Efficiency Metrics ---

Mean inference time: 9.1296 s

Maximum inference time: 10.3325 s

Mean peak GPU memory: 2010.11 MB

Maximum peak GPU memory: 2040.05 MB

A2

AutoMSC Evaluation

--- Segmentation Metrics ---

Label 1: DSC = 0.9222 +/- 0.0370 | NSD = 0.9645 +/- 0.0249

Overall: DSC = 0.9222 +/- 0.0370 | NSD = 0.9645 +/- 0.0249

--- Classification Metrics ---

Accuracy: 0.9167

Balanced Accuracy: 0.5000

AUC: 0.3434

AUPRC: 0.0809

Sensitivity: 0.0000

Specificity: 1.0000

Precision: 0.0000

Recall: 0.0000

F1: 0.0000

Confusion Matrix:

[[33 0]

[3 0]]

--- Balanced Accuracy Verification ---

Definition: unweighted mean of per-class recall

class 0: recall=1.0000, n=33 | class 1: recall=0.0000, n=3

Verified Balanced Accuracy: 0.5000

--- Efficiency Metrics ---

Mean inference time: 9.1393 s

Maximum inference time: 10.3580 s

Mean peak GPU memory: 2401.33 MB

Maximum peak GPU memory: 2431.86 MB

A3

AutoMSC Evaluation

--- Segmentation Metrics ---

Label 1: DSC = 0.9222 +/- 0.0370 | NSD = 0.9645 +/- 0.0249

Overall: DSC = 0.9222 +/- 0.0370 | NSD = 0.9645 +/- 0.0249

--- Classification Metrics ---

Accuracy: 0.3333

Balanced Accuracy: 0.6364

AUC: 0.3434

AUPRC: 0.0829

Sensitivity: 1.0000

Specificity: 0.2727

Precision: 0.1111

Recall: 1.0000

F1: 0.2000

Confusion Matrix:

[[9 24]

[0 3]]

--- Balanced Accuracy Verification ---

Definition: unweighted mean of per-class recall

class 0: recall=0.2727, n=33 | class 1: recall=1.0000, n=3

Verified Balanced Accuracy: 0.6364

--- Efficiency Metrics ---

Mean inference time: 9.1254 s

Maximum inference time: 10.3366 s

Mean peak GPU memory: 2794.56 MB

Maximum peak GPU memory: 2825.36 MB

A4

AutoMSC Evaluation

--- Segmentation Metrics ---

Label 1: DSC = 0.8325 +/- 0.2306 | NSD = 0.8675 +/- 0.2452

Overall: DSC = 0.8325 +/- 0.2306 | NSD = 0.8675 +/- 0.2452

--- Classification Metrics ---

Accuracy: 0.9167

Balanced Accuracy: 0.5000

AUC: 0.8788

AUPRC: 0.5278

Sensitivity: 0.0000

Specificity: 1.0000

Precision: 0.0000

Recall: 0.0000

F1: 0.0000

Confusion Matrix:

[[33 0]

[3 0]]

--- Balanced Accuracy Verification ---

Definition: unweighted mean of per-class recall

class 0: recall=1.0000, n=33 | class 1: recall=0.0000, n=3

Verified Balanced Accuracy: 0.5000

--- Efficiency Metrics ---

Mean inference time: 9.7621 s

Maximum inference time: 26.0649 s

Mean peak GPU memory: 5640.09 MB

Maximum peak GPU memory: 5670.87 MB

A5

AutoMSC Evaluation

--- Segmentation Metrics ---

Label 1: DSC = 0.8325 +/- 0.2306 | NSD = 0.8675 +/- 0.2452

Overall: DSC = 0.8325 +/- 0.2306 | NSD = 0.8675 +/- 0.2452

--- Classification Metrics ---

Accuracy: 0.9167

Balanced Accuracy: 0.5000

AUC: 0.8788

AUPRC: 0.5278

Sensitivity: 0.0000

Specificity: 1.0000

Precision: 0.0000

Recall: 0.0000

F1: 0.0000

Confusion Matrix:

[[33 0]

[3 0]]

--- Balanced Accuracy Verification ---

Definition: unweighted mean of per-class recall

class 0: recall=1.0000, n=33 | class 1: recall=0.0000, n=3

Verified Balanced Accuracy: 0.5000

--- Efficiency Metrics ---

Mean inference time: 9.7621 s

Maximum inference time: 26.0649 s

Mean peak GPU memory: 5640.09 MB

Maximum peak GPU memory: 5670.87 MB

Dataset004_PICAI

A0

AutoMSC Evaluation

--- Segmentation Metrics ---

Label 1: DSC = 0.9587 +/- 0.0346 | NSD = 0.9212 +/- 0.0656

Label 2: DSC = 0.2018 +/- 0.3037 | NSD = 0.2102 +/- 0.3169

Overall: DSC = 0.6253 +/- 0.4274 | NSD = 0.6080 +/- 0.4139

--- Classification Metrics ---

Accuracy: 0.4352

Balanced Accuracy: 0.2985

Balanced Accuracy: 0.2985

Weighted AUC: 0.6738

Weighted AUPRC: 0.4995

Weighted F1: 0.4561

Weighted Precision: 0.5032

Weighted Recall: 0.4352

Per-class metrics:

Class 0: Precision=0.7687, Recall=0.6023, F1=0.6754, AUC=0.7398, AUPRC=0.7515

Class 1: Precision=0.0833, Recall=0.0455, F1=0.0588, AUC=0.4531, AUPRC=0.1378

Class 2: Precision=0.2456, Recall=0.2979, F1=0.2692, AUC=0.6406, AUPRC=0.2493

Class 3: Precision=0.1765, Recall=0.3000, F1=0.2222, AUC=0.6333, AUPRC=0.1223

Class 4: Precision=0.0000, Recall=0.0000, F1=0.0000, AUC=0.6480, AUPRC=0.0748

Class 5: Precision=0.1154, Recall=0.5455, F1=0.1905, AUC=0.7655, AUPRC=0.0920

Confusion Matrix:

[[103 17 24 9 0 18]

[15 2 13 8 0 6]

[12 3 14 7 0 11]

[3 1 2 6 0 8]

[1 0 3 1 0 3]

[0 1 1 3 0 6]]

--- Efficiency Metrics ---

Mean inference time: 13.5351 s

Maximum inference time: 52.3249 s

Mean peak GPU memory: 1832.50 MB

Maximum peak GPU memory: 2022.06 MB

A1

AutoMSC Evaluation

--- Segmentation Metrics ---

Label 1: DSC = 0.9587 +/- 0.0346 | NSD = 0.9212 +/- 0.0656

Label 2: DSC = 0.2018 +/- 0.3037 | NSD = 0.2102 +/- 0.3169

Overall: DSC = 0.6253 +/- 0.4274 | NSD = 0.6080 +/- 0.4139

--- Classification Metrics ---

Accuracy: 0.4983

Balanced Accuracy: 0.3064

Balanced Accuracy: 0.3064

Weighted AUC: 0.7383

Weighted AUPRC: 0.5611

Weighted F1: 0.5026

Weighted Precision: 0.5232

Weighted Recall: 0.4983

Per-class metrics:

Class 0: Precision=0.8217, Recall=0.7544, F1=0.7866, AUC=0.8083, AUPRC=0.8301

Class 1: Precision=0.2812, Recall=0.2045, F1=0.2368, AUC=0.5781, AUPRC=0.2321

Class 2: Precision=0.0000, Recall=0.0000, F1=0.0000, AUC=0.6771, AUPRC=0.2428

Class 3: Precision=0.1429, Recall=0.3000, F1=0.1935, AUC=0.7292, AUPRC=0.2058

Class 4: Precision=0.1000, Recall=0.1250, F1=0.1111, AUC=0.6651, AUPRC=0.0601

Class 5: Precision=0.0862, Recall=0.4545, F1=0.1449, AUC=0.6229, AUPRC=0.0666

Confusion Matrix:

[[129 16 1 6 4 15]

[13 9 1 7 1 13]

[12 4 0 16 0 15]

[3 1 0 6 2 8]

[0 1 0 4 1 2]

[0 1 0 3 2 5]]

--- Efficiency Metrics ---

Mean inference time: 30.2407 s

Maximum inference time: 81.9306 s

Mean peak GPU memory: 1444.11 MB

Maximum peak GPU memory: 2670.37 MB

A2

AutoMSC Evaluation

--- Segmentation Metrics ---

Label 1: DSC = 0.9587 +/- 0.0346 | NSD = 0.9212 +/- 0.0656

Label 2: DSC = 0.2018 +/- 0.3037 | NSD = 0.2102 +/- 0.3169

Overall: DSC = 0.6253 +/- 0.4274 | NSD = 0.6080 +/- 0.4139

--- Classification Metrics ---

Accuracy: 0.5349

Balanced Accuracy: 0.2907

Balanced Accuracy: 0.2907

Weighted AUC: 0.7370

Weighted AUPRC: 0.5423

Weighted F1: 0.5275

Weighted Precision: 0.5934

Weighted Recall: 0.5349

Per-class metrics:

Class 0: Precision=0.8113, Recall=0.7544, F1=0.7818, AUC=0.7891, AUPRC=0.7933

Class 1: Precision=0.5000, Recall=0.0455, F1=0.0833, AUC=0.5771, AUPRC=0.2407

Class 2: Precision=0.2989, Recall=0.5532, F1=0.3881, AUC=0.6855, AUPRC=0.2518

Class 3: Precision=0.1250, Recall=0.0500, F1=0.0714, AUC=0.7231, AUPRC=0.1715

Class 4: Precision=0.0690, Recall=0.2500, F1=0.1081, AUC=0.7568, AUPRC=0.0886

Class 5: Precision=0.0714, Recall=0.0909, F1=0.0800, AUC=0.7991, AUPRC=0.0913

Confusion Matrix:

[[129 1 24 2 12 3]

[15 2 20 3 2 2]

[12 0 26 1 6 2]

[3 0 9 1 3 4]

[0 1 3 0 2 2]

[0 0 5 1 4 1]]

--- Efficiency Metrics ---

Mean inference time: 23.1579 s

Maximum inference time: 55.9165 s

Mean peak GPU memory: 1979.45 MB

Maximum peak GPU memory: 2093.28 MB

A3

AutoMSC Evaluation

--- Segmentation Metrics ---

Label 1: DSC = 0.9587 +/- 0.0346 | NSD = 0.9212 +/- 0.0656

Label 2: DSC = 0.2018 +/- 0.3037 | NSD = 0.2102 +/- 0.3169

Overall: DSC = 0.6253 +/- 0.4274 | NSD = 0.6080 +/- 0.4139

--- Classification Metrics ---

Accuracy: 0.4585

Balanced Accuracy: 0.3149

Balanced Accuracy: 0.3149

Weighted AUC: 0.7248

Weighted AUPRC: 0.5495

Weighted F1: 0.4795

Weighted Precision: 0.6081

Weighted Recall: 0.4585

Per-class metrics:

Class 0: Precision=0.8095, Recall=0.6959, F1=0.7484, AUC=0.7989, AUPRC=0.8126

Class 1: Precision=0.6667, Recall=0.0455, F1=0.0851, AUC=0.5386, AUPRC=0.2418

Class 2: Precision=0.2609, Recall=0.1277, F1=0.1714, AUC=0.6649, AUPRC=0.2478

Class 3: Precision=0.0556, Recall=0.1000, F1=0.0714, AUC=0.6763, AUPRC=0.1385

Class 4: Precision=0.1111, Recall=0.3750, F1=0.1714, AUC=0.7257, AUPRC=0.0892

Class 5: Precision=0.0923, Recall=0.5455, F1=0.1579, AUC=0.6614, AUPRC=0.0621

Confusion Matrix:

[[119 1 7 11 8 25]

[15 2 6 8 1 12]

[10 0 6 11 8 12]

[3 0 3 2 4 8]

[0 0 1 2 3 2]

[0 0 0 2 3 6]]

--- Efficiency Metrics ---

Mean inference time: 23.3278 s

Maximum inference time: 57.2624 s

Mean peak GPU memory: 3070.07 MB

Maximum peak GPU memory: 3185.04 MB

A4

AutoMSC Evaluation

--- Segmentation Metrics ---

Label 1: DSC = 0.9557 +/- 0.0379 | NSD = 0.9120 +/- 0.0707

Label 2: DSC = 0.1667 +/- 0.2810 | NSD = 0.1719 +/- 0.2871

Overall: DSC = 0.5900 +/- 0.4385 | NSD = 0.5690 +/- 0.4210

--- Classification Metrics ---

Accuracy: 0.6246

Balanced Accuracy: 0.3223

Balanced Accuracy: 0.3223

Weighted AUC: 0.8122

Weighted AUPRC: 0.5996

Weighted F1: 0.6079

Weighted Precision: 0.5942

Weighted Recall: 0.6246

Per-class metrics:

Class 0: Precision=0.8192, Recall=0.8480, F1=0.8333, AUC=0.8502, AUPRC=0.8513

Class 1: Precision=0.3393, Recall=0.4318, F1=0.3800, AUC=0.6908, AUPRC=0.2793

Class 2: Precision=0.3654, Recall=0.4043, F1=0.3838, AUC=0.7746, AUPRC=0.3389

Class 3: Precision=0.3333, Recall=0.2500, F1=0.2857, AUC=0.7769, AUPRC=0.2004

Class 4: Precision=0.0000, Recall=0.0000, F1=0.0000, AUC=0.8814, AUPRC=0.1321

Class 5: Precision=0.0000, Recall=0.0000, F1=0.0000, AUC=0.8828, AUPRC=0.1485

Confusion Matrix:

[[145 14 6 6 0 0]

[19 19 5 1 0 0]

[10 16 19 1 0 1]

[3 5 7 5 0 0]

[0 1 5 2 0 0]

[0 1 10 0 0 0]]

--- Efficiency Metrics ---

Mean inference time: 16.0922 s

Maximum inference time: 35.9264 s

Mean peak GPU memory: 4713.12 MB

Maximum peak GPU memory: 4870.11 MB

A5

AutoMSC Evaluation

--- Segmentation Metrics ---

Label 1: DSC = 0.9592 +/- 0.0349 | NSD = 0.9232 +/- 0.0642

Label 2: DSC = 0.2054 +/- 0.3059 | NSD = 0.2141 +/- 0.3186

Overall: DSC = 0.6295 +/- 0.4261 | NSD = 0.6131 +/- 0.4130

--- Classification Metrics ---

Accuracy: 0.5914

Balanced Accuracy: 0.3025

Balanced Accuracy: 0.3025

Weighted AUC: 0.8067

Weighted AUPRC: 0.5977

Weighted F1: 0.5842

Weighted Precision: 0.5891

Weighted Recall: 0.5914

Per-class metrics:

Class 0: Precision=0.8333, Recall=0.7895, F1=0.8108, AUC=0.8431, AUPRC=0.8453

Class 1: Precision=0.2787, Recall=0.3864, F1=0.3238, AUC=0.6783, AUPRC=0.2800

Class 2: Precision=0.3382, Recall=0.4894, F1=0.4000, AUC=0.7677, AUPRC=0.3424

Class 3: Precision=0.3333, Recall=0.1500, F1=0.2069, AUC=0.7950, AUPRC=0.2018

Class 4: Precision=0.0000, Recall=0.0000, F1=0.0000, AUC=0.8814, AUPRC=0.1439

Class 5: Precision=0.0000, Recall=0.0000, F1=0.0000, AUC=0.8890, AUPRC=0.1614

Confusion Matrix:

[[135 25 8 3 0 0]

[17 17 10 0 0 0]

[7 15 23 1 0 1]

[3 3 11 3 0 0]

[0 0 6 2 0 0]

[0 1 10 0 0 0]]

--- Efficiency Metrics ---

Mean inference time: 10.0006 s

Maximum inference time: 35.9264 s

Mean peak GPU memory: 4710.79 MB

Maximum peak GPU memory: 4824.58 MB

Dataset005_LUNA25

A0

AutoMSC Evaluation

--- Segmentation Metrics ---

Label 1: DSC = 0.7722 +/- 0.1557 | NSD = 0.8621 +/- 0.1795

Overall: DSC = 0.7722 +/- 0.1557 | NSD = 0.8621 +/- 0.1795

--- Classification Metrics ---

Accuracy: 0.7835

Balanced Accuracy: 0.7580

AUC: 0.8099

AUPRC: 0.3135

Sensitivity: 0.7264

Specificity: 0.7895

Precision: 0.2655

Recall: 0.7264

F1: 0.3889

Confusion Matrix:

[[799 213]

[29 77]]

--- Balanced Accuracy Verification ---

Definition: unweighted mean of per-class recall

class 0: recall=0.7895, n=1012 | class 1: recall=0.7264, n=106

Verified Balanced Accuracy: 0.7580

--- Efficiency Metrics ---

Mean inference time: 12.1596 s

Maximum inference time: 34.4734 s

Mean peak GPU memory: 3628.00 MB

Maximum peak GPU memory: 4656.34 MB

A1

AutoMSC Evaluation

--- Segmentation Metrics ---

Label 1: DSC = 0.7722 +/- 0.1557 | NSD = 0.8621 +/- 0.1795

Overall: DSC = 0.7722 +/- 0.1557 | NSD = 0.8621 +/- 0.1795

--- Classification Metrics ---

Accuracy: 0.7263

Balanced Accuracy: 0.7812

AUC: 0.8312

AUPRC: 0.3466

Sensitivity: 0.8491

Specificity: 0.7134

Precision: 0.2368

Recall: 0.8491

F1: 0.3704

Confusion Matrix:

[[722 290]

[16 90]]

--- Balanced Accuracy Verification ---

Definition: unweighted mean of per-class recall

class 0: recall=0.7134, n=1012 | class 1: recall=0.8491, n=106

Verified Balanced Accuracy: 0.7812

--- Efficiency Metrics ---

Mean inference time: 8.0005 s

Maximum inference time: 10.4547 s

Mean peak GPU memory: 3628.26 MB

Maximum peak GPU memory: 3645.15 MB

A2

AutoMSC Evaluation

--- Segmentation Metrics ---

Label 1: DSC = 0.7722 +/- 0.1557 | NSD = 0.8621 +/- 0.1795

Overall: DSC = 0.7722 +/- 0.1557 | NSD = 0.8621 +/- 0.1795

--- Classification Metrics ---

Accuracy: 0.7227

Balanced Accuracy: 0.7750

AUC: 0.8259

AUPRC: 0.3671

Sensitivity: 0.8396

Specificity: 0.7105

Precision: 0.2330

Recall: 0.8396

F1: 0.3648

Confusion Matrix:

[[719 293]

[17 89]]

--- Balanced Accuracy Verification ---

Definition: unweighted mean of per-class recall

class 0: recall=0.7105, n=1012 | class 1: recall=0.8396, n=106

Verified Balanced Accuracy: 0.7750

--- Efficiency Metrics ---

Mean inference time: 7.9500 s

Maximum inference time: 10.4883 s

Mean peak GPU memory: 3629.67 MB

Maximum peak GPU memory: 3646.70 MB

A3

AutoMSC Evaluation

--- Segmentation Metrics ---

Label 1: DSC = 0.7722 +/- 0.1557 | NSD = 0.8621 +/- 0.1795

Overall: DSC = 0.7722 +/- 0.1557 | NSD = 0.8621 +/- 0.1795

--- Classification Metrics ---

Accuracy: 0.7692

Balanced Accuracy: 0.7923

AUC: 0.8422

AUPRC: 0.3681

Sensitivity: 0.8208

Specificity: 0.7638

Precision: 0.2669

Recall: 0.8208

F1: 0.4028

Confusion Matrix:

[[773 239]

[19 87]]

--- Balanced Accuracy Verification ---

Definition: unweighted mean of per-class recall

class 0: recall=0.7638, n=1012 | class 1: recall=0.8208, n=106

Verified Balanced Accuracy: 0.7923

--- Efficiency Metrics ---

Mean inference time: 16.6237 s

Maximum inference time: 87.6727 s

Mean peak GPU memory: 1010.21 MB

Maximum peak GPU memory: 2038.82 MB

A4

AutoMSC Evaluation

--- Segmentation Metrics ---

Label 1: DSC = 0.7422 +/- 0.2108 | NSD = 0.8274 +/- 0.2421

Overall: DSC = 0.7422 +/- 0.2108 | NSD = 0.8274 +/- 0.2421

--- Classification Metrics ---

Accuracy: 0.8506

Balanced Accuracy: 0.7021

AUC: 0.8282

AUPRC: 0.3652

Sensitivity: 0.5189

Specificity: 0.8854

Precision: 0.3216

Recall: 0.5189

F1: 0.3971

Confusion Matrix:

[[896 116]

[51 55]]

--- Balanced Accuracy Verification ---

Definition: unweighted mean of per-class recall

class 0: recall=0.8854, n=1012 | class 1: recall=0.5189, n=106

Verified Balanced Accuracy: 0.7021

--- Efficiency Metrics ---

Mean inference time: 7.5634 s

Maximum inference time: 44.9141 s

Mean peak GPU memory: 3621.71 MB

Maximum peak GPU memory: 3639.06 MB

A5

AutoMSC Evaluation

--- Segmentation Metrics ---

Label 1: DSC = 0.7837 +/- 0.1431 | NSD = 0.8685 +/- 0.1685

Overall: DSC = 0.7837 +/- 0.1431 | NSD = 0.8685 +/- 0.1685

--- Classification Metrics ---

Accuracy: 0.8712

Balanced Accuracy: 0.7177

AUC: 0.8421

AUPRC: 0.4038

Sensitivity: 0.5283

Specificity: 0.9071

Precision: 0.3733

Recall: 0.5283

F1: 0.4375

Confusion Matrix:

[[918 94]

[50 56]]

--- Balanced Accuracy Verification ---

Definition: unweighted mean of per-class recall

class 0: recall=0.9071, n=1012 | class 1: recall=0.5283, n=106

Verified Balanced Accuracy: 0.7177

--- Efficiency Metrics ---

Mean inference time: 8.7610 s

Maximum inference time: 39.0454 s

Mean peak GPU memory: 2944.32 MB

Maximum peak GPU memory: 3108.95 MB

Dataset061_PETWB_Lung

A0

AutoMSC Evaluation

--- Segmentation Metrics ---

Label 1: DSC = 0.9879 +/- 0.0033 | NSD = 0.9413 +/- 0.0135

Overall: DSC = 0.9879 +/- 0.0033 | NSD = 0.9413 +/- 0.0135

--- Classification Metrics ---

Accuracy: 0.9082

Balanced Accuracy: 0.6653

AUC: 0.6542

AUPRC: 0.9575

Sensitivity: 0.9556

Specificity: 0.3750

Precision: 0.9451

Recall: 0.9556

F1: 0.9503

Confusion Matrix:

```
[[ 3  5]
 [ 4 86]]
```

--- Balanced Accuracy Verification ---

Definition: unweighted mean of per-class recall

class 0: recall=0.3750, n=8 | class 1: recall=0.9556, n=90

Verified Balanced Accuracy: 0.6653

--- Efficiency Metrics ---

Mean inference time: 7.8818 s

Maximum inference time: 29.3788 s

Mean peak GPU memory: 2279.75 MB

Maximum peak GPU memory: 3265.54 MB

A1

AutoMSC Evaluation

--- Segmentation Metrics ---

Label 1: DSC = 0.9879 +/- 0.0033 | NSD = 0.9413 +/- 0.0135

Overall: DSC = 0.9879 +/- 0.0033 | NSD = 0.9413 +/- 0.0135

--- Classification Metrics ---

Accuracy: 0.6531

Balanced Accuracy: 0.6403

AUC: 0.6556

AUPRC: 0.9575

Sensitivity: 0.6556

Specificity: 0.6250

Precision: 0.9516

Recall: 0.6556

F1: 0.7763

Confusion Matrix:

[[5 3]

[31 59]]

--- Balanced Accuracy Verification ---

Definition: unweighted mean of per-class recall

class 0: recall=0.6250, n=8 | class 1: recall=0.6556, n=90

Verified Balanced Accuracy: 0.6403

--- Efficiency Metrics ---

Mean inference time: 3.8279 s

Maximum inference time: 5.0724 s

Mean peak GPU memory: 2663.23 MB

Maximum peak GPU memory: 2713.48 MB

A2

AutoMSC Evaluation

--- Segmentation Metrics ---

Label 1: DSC = 0.9879 +/- 0.0033 | NSD = 0.9413 +/- 0.0135

Overall: DSC = 0.9879 +/- 0.0033 | NSD = 0.9413 +/- 0.0135

--- Classification Metrics ---

Accuracy: 0.7449

Balanced Accuracy: 0.7472

AUC: 0.7417

AUPRC: 0.9670

Sensitivity: 0.7444

Specificity: 0.7500

Precision: 0.9710

Recall: 0.7444

F1: 0.8428

Confusion Matrix:

[[6 2]

[23 67]]

--- Balanced Accuracy Verification ---

Definition: unweighted mean of per-class recall

class 0: recall=0.7500, n=8 | class 1: recall=0.7444, n=90

Verified Balanced Accuracy: 0.7472

--- Efficiency Metrics ---

Mean inference time: 3.8902 s

Maximum inference time: 5.2628 s

Mean peak GPU memory: 3056.12 MB

Maximum peak GPU memory: 3106.72 MB

A3

AutoMSC Evaluation

--- Segmentation Metrics ---

Label 1: DSC = 0.9879 +/- 0.0033 | NSD = 0.9413 +/- 0.0135

Overall: DSC = 0.9879 +/- 0.0033 | NSD = 0.9413 +/- 0.0135

--- Classification Metrics ---

Accuracy: 0.5612

Balanced Accuracy: 0.6472

AUC: 0.6569

AUPRC: 0.9567

Sensitivity: 0.5444

Specificity: 0.7500

Precision: 0.9608

Recall: 0.5444

F1: 0.6950

Confusion Matrix:

[[6 2]

[41 49]]

--- Balanced Accuracy Verification ---

Definition: unweighted mean of per-class recall

class 0: recall=0.7500, n=8 | class 1: recall=0.5444, n=90

Verified Balanced Accuracy: 0.6472

--- Efficiency Metrics ---

Mean inference time: 3.8729 s

Maximum inference time: 5.0973 s

Mean peak GPU memory: 3448.94 MB

Maximum peak GPU memory: 3499.12 MB

A4

AutoMSC Evaluation

--- Segmentation Metrics ---

Label 1: DSC = 0.9879 +/- 0.0033 | NSD = 0.9413 +/- 0.0135

Overall: DSC = 0.9879 +/- 0.0033 | NSD = 0.9413 +/- 0.0135

--- Classification Metrics ---

Accuracy: 0.9184

Balanced Accuracy: 0.5000

AUC: 0.6556

AUPRC: 0.9586

Sensitivity: 1.0000

Specificity: 0.0000

Precision: 0.9184

Recall: 1.0000

F1: 0.9574

Confusion Matrix:

[[0 8]

[0 90]]

--- Balanced Accuracy Verification ---

Definition: unweighted mean of per-class recall

class 0: recall=0.0000, n=8 | class 1: recall=1.0000, n=90

Verified Balanced Accuracy: 0.5000

--- Efficiency Metrics ---

Mean inference time: 3.9267 s

Maximum inference time: 5.3088 s

Mean peak GPU memory: 5687.38 MB

Maximum peak GPU memory: 5737.98 MB

A5

AutoMSC Evaluation

--- Segmentation Metrics ---

Label 1: DSC = 0.9879 +/- 0.0033 | NSD = 0.9413 +/- 0.0135

Overall: DSC = 0.9879 +/- 0.0033 | NSD = 0.9413 +/- 0.0135

--- Classification Metrics ---

Accuracy: 0.9184

Balanced Accuracy: 0.5000

AUC: 0.6597

AUPRC: 0.9601

Sensitivity: 1.0000

Specificity: 0.0000

Precision: 0.9184

Recall: 1.0000

F1: 0.9574

Confusion Matrix:

```
[[ 0  8]
 [ 0 90]]
```

--- Balanced Accuracy Verification ---

Definition: unweighted mean of per-class recall

class 0: recall=0.0000, n=8 | class 1: recall=1.0000, n=90

Verified Balanced Accuracy: 0.5000

--- Efficiency Metrics ---

Mean inference time: 7.8858 s

Maximum inference time: 28.6464 s

Mean peak GPU memory: 3720.36 MB

Maximum peak GPU memory: 4706.87 MB

Progress 4

A0–A5 Rankings Within Each Training Dataset

Dataset001 — BrainMets

Rank	Method	DSC	NSD	BalAcc	AUROC	Average
1	A4	0.8110	0.9027	0.6538	0.6288	0.7491
2	A5	0.8147	0.9078	0.6488	0.6120	0.7458
3	A2	0.8156	0.9090	0.6104	0.5920	0.7318
4	A3	0.8156	0.9090	0.6321	0.5184	0.7188
5	A1	0.8156	0.9090	0.6321	0.4649	0.7054
6	A0	0.8156	0.9090	0.5000	0.1438	0.5921

Dataset002 — MU-Glioma Post

Rank	Method	DSC	NSD	BalAcc	AUROC	Average
1	A1	0.7900	0.8368	0.6553	0.7818	0.7660
2	A5	0.7908	0.8355	0.5829	0.7837	0.7482
3	A3	0.7900	0.8368	0.6095	0.7408	0.7443
4	A2	0.7900	0.8368	0.6352	0.7044	0.7416
5	A0	0.7900	0.8368	0.5294	0.7497	0.7265
6	A4	0.5939	0.6280	0.6365	0.7955	0.6635

Dataset003 — UCSD-PTGB

Rank	Method	DSC	NSD	BalAcc	AUROC	Average
1	A0	0.9222	0.9645	0.6364	0.6061	0.7823
=2	A4	0.8325	0.8675	0.5000	0.8788	0.7697
=2	A5	0.8325	0.8675	0.5000	0.8788	0.7697
4	A3	0.9222	0.9645	0.6364	0.3434	0.7166
5	A2	0.9222	0.9645	0.5000	0.3434	0.6825
6	A1	0.9222	0.9645	0.4848	0.3131	0.6712

Dataset004 — PI-CAI

Rank	Method	DSC	NSD	BalAcc	AUROC	Average
1	A5	0.6295	0.6131	0.3025	0.8067	0.5879
2	A4	0.5900	0.5690	0.3223	0.8122	0.5734
3	A1	0.6253	0.6080	0.3064	0.7383	0.5695
4	A3	0.6253	0.6080	0.3149	0.7248	0.5683
5	A2	0.6253	0.6080	0.2907	0.7370	0.5653
6	A0	0.6253	0.6080	0.2985	0.6738	0.5514

Dataset005 — LUNA25

Rank	Method	DSC	NSD	BalAcc	AUROC	Average
1	A3	0.7722	0.8621	0.7923	0.8422	0.8172
2	A1	0.7722	0.8621	0.7812	0.8312	0.8117
3	A2	0.7722	0.8621	0.7750	0.8259	0.8088
4	A5	0.7837	0.8685	0.7177	0.8421	0.8030
5	A0	0.7722	0.8621	0.7580	0.8099	0.8005
6	A4	0.7422	0.8274	0.7021	0.8282	0.7750

Dataset061 — PETWB Lung

Rank	Method	DSC	NSD	BalAcc	AUROC	Average
1	A2	0.9879	0.9413	0.7472	0.7417	0.8545
2	A0	0.9879	0.9413	0.6653	0.6542	0.8122
3	A3	0.9879	0.9413	0.6472	0.6569	0.8083
4	A1	0.9879	0.9413	0.6403	0.6556	0.8063
5	A5	0.9879	0.9413	0.5000	0.6597	0.7722
6	A4	0.9879	0.9413	0.5000	0.6556	0.7712

Dataset062 — PETWB Liver

Rank	Method	DSC	NSD	BalAcc	AUROC	Average
1	A1	0.9786	0.9006	0.5918	0.6031	0.7685
2	A3	0.9786	0.9006	0.6020	0.5706	0.7630
3	A5	0.9786	0.9006	0.5816	0.5494	0.7526
4	A2	0.9786	0.9006	0.5714	0.5589	0.7524
5	A4	0.9786	0.9006	0.5816*	0.5410	0.7505
6	A0	0.9786	0.9006	0.5612	0.5352	0.7439

Dataset062_PETWB_Liver

A0

AutoMSC Evaluation

--- Segmentation Metrics ---

Label 1: DSC = 0.9786 +/- 0.0064 | NSD = 0.9006 +/- 0.0220

Overall: DSC = 0.9786 +/- 0.0064 | NSD = 0.9006 +/- 0.0220

--- Classification Metrics ---

Accuracy: 0.5612

Balanced Accuracy: 0.5612

AUC: 0.5352

AUPRC: 0.5745

Sensitivity: 0.9796

Specificity: 0.1429

Precision: 0.5333

Recall: 0.9796

F1: 0.6906

Confusion Matrix:

[[7 42]

[1 48]]

--- Balanced Accuracy Verification ---

Definition: unweighted mean of per-class recall

class 0: recall=0.1429, n=49 | class 1: recall=0.9796, n=49

Verified Balanced Accuracy: 0.5612

--- Efficiency Metrics ---

Mean inference time: 4.9975 s

Maximum inference time: 31.6073 s

Mean peak GPU memory: 2426.97 MB

Maximum peak GPU memory: 3426.19 MB

A1

AutoMSC Evaluation

--- Segmentation Metrics ---

Label 1: DSC = 0.9786 +/- 0.0064 | NSD = 0.9006 +/- 0.0220

Overall: DSC = 0.9786 +/- 0.0064 | NSD = 0.9006 +/- 0.0220

--- Classification Metrics ---

Accuracy: 0.5918

Balanced Accuracy: 0.5918

AUC: 0.6031

AUPRC: 0.6613

Sensitivity: 0.5102

Specificity: 0.6735

Precision: 0.6098

Recall: 0.5102

F1: 0.5556

Confusion Matrix:

[[33 16]

[24 25]]

--- Balanced Accuracy Verification ---

Definition: unweighted mean of per-class recall

class 0: recall=0.6735, n=49 | class 1: recall=0.5102, n=49

Verified Balanced Accuracy: 0.5918

--- Efficiency Metrics ---

Mean inference time: 4.1858 s

Maximum inference time: 5.5583 s

Mean peak GPU memory: 2809.43 MB

Maximum peak GPU memory: 2846.07 MB

A2

AutoMSC Evaluation

--- Segmentation Metrics ---

Label 1: DSC = 0.9786 +/- 0.0064 | NSD = 0.9006 +/- 0.0220

Overall: DSC = 0.9786 +/- 0.0064 | NSD = 0.9006 +/- 0.0220

--- Classification Metrics ---

Accuracy: 0.5714

Balanced Accuracy: 0.5714

AUC: 0.5589

AUPRC: 0.6021

Sensitivity: 0.7347

Specificity: 0.4082

Precision: 0.5538

Recall: 0.7347

F1: 0.6316

Confusion Matrix:

[[20 29]

[13 36]]

--- Balanced Accuracy Verification ---

Definition: unweighted mean of per-class recall

class 0: recall=0.4082, n=49 | class 1: recall=0.7347, n=49

Verified Balanced Accuracy: 0.5714

--- Efficiency Metrics ---

Mean inference time: 4.1529 s

Maximum inference time: 5.3532 s

Mean peak GPU memory: 3201.74 MB

Maximum peak GPU memory: 3238.52 MB

A3

AutoMSC Evaluation

--- Segmentation Metrics ---

Label 1: DSC = 0.9786 +/- 0.0064 | NSD = 0.9006 +/- 0.0220

Overall: DSC = 0.9786 +/- 0.0064 | NSD = 0.9006 +/- 0.0220

--- Classification Metrics ---

Accuracy: 0.6020

Balanced Accuracy: 0.6020

AUC: 0.5706

AUPRC: 0.6119

Sensitivity: 0.6735

Specificity: 0.5306

Precision: 0.5893

Recall: 0.6735

F1: 0.6286

Confusion Matrix:

[[26 23]

[16 33]]

--- Balanced Accuracy Verification ---

Definition: unweighted mean of per-class recall

class 0: recall=0.5306, n=49 | class 1: recall=0.6735, n=49

Verified Balanced Accuracy: 0.6020

--- Efficiency Metrics ---

Mean inference time: 4.1603 s

Maximum inference time: 5.6301 s

Mean peak GPU memory: 3595.72 MB

Maximum peak GPU memory: 3631.42 MB

A4

AutoMSC Evaluation

--- Segmentation Metrics ---

Label 1: DSC = 0.9786 +/- 0.0064 | NSD = 0.9006 +/- 0.0220

Overall: DSC = 0.9786 +/- 0.0064 | NSD = 0.9006 +/- 0.0220

--- Classification Metrics ---

Accuracy: 0.5816

AUC: 0.5410

AUPRC: 0.5716

Sensitivity: 1.0000

Specificity: 0.1633

Precision: 0.5444

Recall: 1.0000

F1: 0.7050

Confusion Matrix:

[[8 41]

[0 49]]

--- Efficiency Metrics ---

Mean inference time: 3.9421 s

Maximum inference time: 9.0003 s

Mean peak GPU memory: 3326.11 MB

Maximum peak GPU memory: 3361.84 MB

A5

AutoMSC Evaluation

--- Segmentation Metrics ---

Label 1: DSC = 0.9786 +/- 0.0064 | NSD = 0.9006 +/- 0.0220

Overall: DSC = 0.9786 +/- 0.0064 | NSD = 0.9006 +/- 0.0220

--- Classification Metrics ---

Accuracy: 0.5816

Balanced Accuracy: 0.5816

AUC: 0.5494

AUPRC: 0.6118

Sensitivity: 0.1837

Specificity: 0.9796

Precision: 0.9000

Recall: 0.1837

F1: 0.3051

Confusion Matrix:

[[48 1]

[40 9]]

--- Balanced Accuracy Verification ---

Definition: unweighted mean of per-class recall

class 0: recall=0.9796, n=49 | class 1: recall=0.1837, n=49

Verified Balanced Accuracy: 0.5816

--- Efficiency Metrics ---

Mean inference time: 13.5857 s

Maximum inference time: 831.3091 s

Mean peak GPU memory: 3475.35 MB

Maximum peak GPU memory: 3511.51 MB